

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		A FeCl ₃ (1) B NaCl (1) C Br ₂ (1) award (1) for correct names of all three substances		3		3		
		(ii)		<u>fume cupboard</u> – because <u>chlorine / bromine / halogen</u> fumes are <u>toxic</u> three parts needed neutral: reference to goggles or general laboratory safety or fumes in general	1			1		1
		(iii)		no reaction / no change / no observable change (1) iodine is less reactive than bromine (1) and is therefore unable to <u>displace the bromide / bromine</u> (from the solution/sodium bromide) (1) neutral: iodine less reactive than chlorine gains no credit	2	1		3		3

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			ClO_2 (3) if formula incorrect award credit for correct steps $Cl = \frac{0.71}{35.5}/0.02$ and $O = \frac{0.64}{16}/0.04$ (1) conversion of 0.02 and 0.04 to a 1:2 ratio (1) ecf possible award (1) max if A_r divided by mass leading to Cl_2O		3		3	3	
	(c)	(i)		$\text{Ag}^+(\text{aq}) + \text{Br}^-(\text{aq}) \rightarrow \text{AgBr}(\text{s})$ correct formulae for both ions and product (1) state symbols for both ions and product (1) state symbols only credited if ions and product correct	1	1		2		
		(ii)		$2\text{AgNO}_3 + \text{CaI}_2 \rightarrow 2\text{AgI} + \text{Ca}(\text{NO}_3)_2$ correct reactants (1) correct products (1) balancing (1) balancing mark can only be awarded if both the reactants and products are correct		3		3	1	
				Question 5 total	4	11	0	15	4	4